

Rare-earth residue prediction

E1-Bind training, AlphaFold 3 structural validation, data provenance, and complete held-out results

Technical evidence report • 12 September 2026 (EST) • Current run plus clearly marked historical repository evidence

Main outcome. The September held-out E1-Bind model produced residue-level REE probabilities for 37 distinct test chains. AF3 APO and rare-earth HOLO models completed for all 25 test chains that have at least one positive reference residue. On those 25 selected chains, intersecting E1-Bind positives with AF3 ion contacts raised matched-subtype micro-F1 from 0.084 to 0.313; for the combined REE output, from 0.069 to 0.297. This is a diagnostic positive-only subset result, not an all-37 test score or experimental proof of binding.

1. Scope and status

This document consolidates the E1-Bind/rare-earth/AF3 work discussed in this project. It distinguishes (a) historical results already documented in the E1-Metal-Binding repository from (b) the September 2026 run executed on the local compute node. It does not summarize unrelated protein-design benchmark experiments elsewhere in the workspace. Exact machine-readable results remain in `summary.json`, `per_sample.csv`, `test_metrics.yaml` and `holo_receipt.tsv` beside this PDF.

Stage	September status	Evidence
AF3/E1 environment and assets	Ready	AF3 databases 627 GB; model weights 2.1 GB; state markers present
REE data/MSAs/embeddings	Ready	186 unique chain IDs; eight-view E1 embeddings; LREE/HREE stores
E1-Bind training	Complete	Five folds; 149 distinct train chains; 14.5 min training log
Held-out residue inference	Complete	37 distinct test chains; LREE 16, HREE 22, one shared
AF3 smoke test	Complete	130-aa protein-only 1CNT_2; pTM 0.65
Held-out APO and HOLO	Complete on selected subset	25/25 APO and 25/25 HOLO model receipts
All-37 AF3-assisted score	Not available	12 all-zero-label test chains were not run through AF3

2. Concepts: protein, residue, label, rare earth

A PDB entry is a deposited 3D structure, not necessarily one unique protein; it may include several protein chains, ions and other molecules. A sample here is a chain ID such as 1EXZ_B, with an amino-acid sequence and one binary label per residue. Protein length therefore varies by sample, and the number of residue-level predictions equals that chain's sequence length. The dataset is not a census of all proteins in the Protein Data Bank. The candidate metadata locally contains 1,244 rows, 373 distinct PDB IDs and 822 distinct PDB-chain pairs before final filtering.

REE means rare-earth elements. The project's **LREE** set is La, Ce, Pr, Nd, Pm and Sm; its **HREE** set is Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu and Y. This is the repository's operational partition; Y is included with HREE. REE is a logical combined prediction target, not a separate embedding store or a physical ion in the AF3 input. A metadata element code 'REE' may denote a nonspecific/negative metadata row, so it should not be mistaken for a chemical element.

3. Original data and candidate dataset

The source workflow starts from experimental structure files and PLIP metal-complex contact reports. It maps protein contacts onto chain sequences, rejects waters and non-amino-acid contacts, constructs positive/negative residue labels, retains a capped number of fully negative chains, clusters homologous sequences, and makes a train/test split. A label is evidence under this contact-extraction convention; a zero is not proof that a residue cannot bind a metal.

The September training used the **May 12 QA rebuild candidate**, not the older tracked April 15 canonical FASTAs. The QA candidate has not been promoted to the tracked canonical files. Its config uses 50–1,500-aa sequences, a 0.2 negative-to-positive retention ratio (seed 42), CD-HIT at 0.5 sequence identity, and an 80/20 split (seed 42)

grouped by exact sequence hash. The source report cites 1,261 PLIP metadata rows before the QA rebuild; the rebuilt candidate metadata file currently has 1,244 rows.

QA contact category	Count
accepted protein	1,486
rejected non aa	332
rejected water	628
unmapped protein	57
unsupported metal	2,187
Total audited contacts	4,690
GD3 alias normalized	85

The candidate-vs-older-dataset audit reported eight more sequences and 24 more binding residues in aggregate, 55 added and 47 removed sequence IDs, 33 moved split IDs, and no changed sequence or label vector for IDs that remained directly comparable.

The rebuilt metadata's element-code composition is shown below. 'REE' is a generic code in these rows, not an elemental species; the final positive structural cohort uses the specific element identified in each experimental CIF.

Element code	Metadata rows	Element code	Metadata rows
CE	6	PR	73
EU	12	REE	844
GD	17	SM	56
LA	29	TB	73
LU	20	YB	114

Subtype / split	Chains	Positive chains	All-zero chains	Residues	Positive residues	Positive fraction
LREE train	82	66	16	20,819	225	1.081%
LREE test	16	9	7	3,496	28	0.801%
HREE train	68	48	20	19,684	144	0.732%
HREE test	22	16	6	9,164	38	0.415%

LREE/HREE row counts are subtype views, not four disjoint protein sets. Two IDs are shared across the full subtype dataset; the distinct-chain union is 186, split into 149 train and 37 test chains. The split contains 178 exact-sequence-hash groups. The test has one shared LREE/HREE chain ID, so 16 + 22 subtype records become 37 distinct test chains. Only 25 of those 37 have at least one positive reference residue in the selected subtype (nine LREE, sixteen HREE); the other 12 are all-zero in the evaluated view.

4. Installation, data preparation and training

The compute pipeline waited for the AF3 Python/GPU environment, official AF3 databases and model weights, and E1/E1-Bind environments. The complete local AF3 database directory occupies about 627 GB; weights about 2.1 GB. The long download was principally a large one-time asset transfer; this report has no trustworthy per-hour bandwidth trace from which to assign a more specific slowdown cause. The running pipeline was organized in tmux sessions so terminal disconnection did not terminate it. A user can detach with Ctrl-b, d and reattach via SSH plus tmux attach -t SESSION, provided the remote host/session still exists.

After assets were ready, MMseqs2 built CPU and GPU-padded UniRef90 databases and searched the 186-chain query FASTA to create A3M MSAs. The Profluent-Bio/E1-600m model generated eight embedding views per sequence. The pipeline checked all 186 embedding files for shape [8, length, 1280] and finite values, then made LREE/HREE NumPy stores. The raw E1 embedding directory occupies about 2.1 GB; the NumPy stores about 2.1 GB. The combined REE arm reuses the physical subtype stores.

The September E1-Bind run started 17:00:02 and finished 17:14:30 on September 12, 2026 (times in the compute-node log). The saved config specifies five-fold cross-validation, E1 residue embeddings, 4 of the 8

generated variants per training example, 256 hidden units, 0.3 dropout, batch size 2,048, stage-1 maximum 10 epochs, stage-2 maximum 30 epochs, early stopping, BF16, and Youden-selected thresholds. Five model checkpoints and OOF predictions were saved; this is trained E1-Bind, not merely a frozen E1 embedding calculation. The fold training/OOF process used the 149 train chains, then performed separate test inference on 37. The reported mean fold AUPRC was 0.014 ± 0.004 ; it is not a held-out-test F1.

Output arm	Threshold	Held-out subtype records
LREE	0.265625	16
HREE	0.27734375	22
REE	0.23046875	37 distinct chains

The per-residue model score is a decision score/probability-like output for a residue's metal-binding label under the trained model. Thresholding turns it into a binary predicted site. It is not a measured binding affinity, a Kd, or AF3's confidence. $F1 = 2TP/(2TP+FP+FN)$; precision = $TP/(TP+FP)$, recall = $TP/(TP+FN)$. Micro metrics pool residue counts across proteins.

5. AF3 smoke test: what it did and did not check

A 130-residue protein-only input named 1CNT_2 was used to confirm AF3 end-to-end installation and inference. Its JSON contains one protein chain A, model seed 1, and no rare-earth ion. The full-data-pipeline run created a model and five seed-1 sample outputs; an inference-only JSON with a query-only MSA was also prepared for a lighter path. The protein-only top summary reported pTM 0.65, ranking score 0.65, and no meaningful inter-chain ipTM. This tests whether AF3 can process that sequence and produce a plausible monomeric fold. It does **not** test rare-earth coordination, binding-site accuracy, or agreement with the experimental PDB structure.

pTM is AF3's predicted template-modeling score: a model-internal estimate of global topology accuracy, on a 0-1 scale. In this smoke input, a 0.65 value is a middling/moderate global-fold confidence indicator. No ground-truth structure was compared by the pTM calculation, and the number is not a probability that each residue or any metal-binding site is correct. A protein-only pTM cannot validate a protein-rare-earth complex that was not in the input.

6. September held-out AF3-assisted validation protocol

The model's test output covered 37 distinct chains. The structural follow-up intentionally selected the 25 chains with at least one positive PLIP-derived label. The exact chain sequence was used as AF3 protein chain A. Experimental mmCIF files were read to identify the actual rare-earth element for each positive case. The 25-case element mix was EU 1, LA 3, LU 2, PR 4, SM 2, TB 8, YB 5. This means the HOLO validation used La/Sm/Pr/Eu/Tb/Yb/Lu as observed, not a generic La substitute for every case. Experimental structure coordinates were not supplied as AF3 input for predicting ion positions; they were used for element identification and label provenance.

First, AF3 generated 25 APO (protein-only) structures. The E1-Bind subtype residue scores and an APO 5.0-Å C β contact graph (alpha 0.2, two smoothing steps, weighted chemistry gate) fed ion-count estimation. Base count was increased by two when positive. HOLO AF3 JSONs then contained the same protein sequence plus that many ions of the case's experimental element. Two model seeds, 1 and 42, were configured; the analysis used the top-ranked output model. Each selected case's predicted chain-A sequence was checked against the submitted sequence before scoring.

The 25 APO jobs all completed; their receipt sums to 16,731 seconds (4 h 38 m 51 s) of sequential job runtime, median 524 s, range 402-2,223 s. The 25 HOLO jobs all completed; their receipt sums to 8,043 seconds (2 h 14 m 03 s), median 186 s, range 80-1,713 s. Combined recorded AF3 job time is 24,774 seconds (6 h 52 m 54 s), excluding other workflow steps and any idle/transfer time. AF3 modeling, especially APO/MSA/template preparation, dominated the 14.5-minute E1 training runtime.

The script's contact rule is geometric: a predicted ion and a protein oxygen, nitrogen, or sulfur atom at most 3.5 Å apart mark the residue as contacting. This is **not PLIP**; the PLIP executable was unavailable to this September analysis. The candidate rules were: raw E1 subtype positives; raw combined REE positives; AF3 contacts alone; and each raw set intersected with AF3 contacts. Labels are the held-out candidate FASTA's binary residue vectors. No score cutoff was chosen from these 25 outcomes.

The 25 selected chains span 60-1,369 amino acids (median 364; 9,721 residues pooled), with 66 labeled positive residues. They requested 348 HOLO ions in total (4-42 per case), much more than 66 labeled contact residues; ion counts and residue counts are distinct units. The selected models' pTM values range 0.60-0.97 (median 0.85). These confidence values were recorded, but not used to filter or calibrate the F1 results.

An intermediate preparation manifest had blank `sample_id` cells because its result object used `sequence_id` instead. The final analysis reader accepts either key; the 25 chain identities were verified against input/output filenames, parsed AF3 chain sequences, and the complete per-sample CSV. The manifest display issue did not change the scored cohort or counts.

7. Full held-out E1-only test, before AF3

These are the September model's saved test metrics. LREE and HREE denominators are their respective subtype records, whereas REE is the 37-distinct-chain combined view. The REE all-37 raw score is the relevant unfiltered baseline for a future all-37 AF3 comparison.

Arm	Chains	TP	FP	FN	Precision	Recall	F1	AUPRC
LREE	16	24	540	4	0.0426	0.8571	0.0811	0.1950
HREE	22	28	903	10	0.0301	0.7368	0.0578	0.1442
REE	37	58	1,984	8	0.0284	0.8788	0.0550	0.1643

The saved test metrics also show REE ROC-AUC 0.918 and AUPRC 0.164. A ranking statistic can look much stronger than thresholded F1 in a dataset with very sparse positives; the clinically or experimentally usable site list still depends on precision and calibration.

8. AF3-assisted result on the selected 25 chains

Rule	TP	FP	FN	TN	Precision	Recall	Micro-F1
Raw matched subtype	52	1,119	14	8,536	0.0444	0.7879	0.0841
AF3 contacts alone	57	275	9	9,380	0.1717	0.8636	0.2864
Subtype n AF3	48	193	18	9,462	0.1992	0.7273	0.3127
Raw combined REE	58	1,550	8	8,105	0.0361	0.8788	0.0693
REE n AF3	51	227	15	9,428	0.1835	0.7727	0.2965

Subtype intersection reduced false positives 1,119 → 193 (−926, 82.8%) but also reduced true positives 52 → 48. Its precision rose 0.0444 → 0.1992; recall fell 0.7879 → 0.7273. For combined REE, false positives fell 1,550 → 227 (−1,323, 85.4%) and true positives 58 → 51. These are within the same 25 selected proteins.

A paired protein bootstrap resampled the 25 rows 10,000 times with seed 42 and recalculated pooled micro-F1 differences. The subtype delta was +0.2286 (95% percentile interval [+0.1799, +0.2822]); combined REE delta +0.2272 (interval [+0.1842, +0.2749]). These intervals describe variation conditional on this selected cohort and protocol; they do not correct the positive-only selection bias.

Selected subgroup	Chains	Subtype raw F1	Subtype assisted F1	REE raw F1	REE assisted F1
LREE	9	0.1191	0.3239	0.1046	0.3129
HREE	16	0.0671	0.3030	0.0560	0.2843

9. How this relates to the historical 0.132 and 0.346 results

The repository documents several earlier, distinct stages. A May 17 retrospective APO contact-prior ablation on 75 overlapping samples changed mean estimated ion count 9.84 → 8.17 and residue micro-F1 0.163 → 0.182; it did not run new HOLO structures. A subsequent May 20 full APO/HOLO pass analyzed 149/149 samples, reusing 85 APO and 10 count-matched HOLO models while generating 64 APO and 139 HOLO models. It requested/found 1,271 La ions, which already motivated ion-count-padding caution. Its internal geometric diagnostic gave raw-ML F1 0.1392 and ML n AF3 F1 0.3448, but PLIP-compatible scoring was designated the authoritative historical comparison.

A previously documented May 2026 **different experiment** used 149 REE out-of-fold protein predictions and AF3 HOLO structures, then compared them to PLIP-compatible contact labels on all 149. It reported raw ML micro-F1 0.1323 (TP 296, FP 3,809, FN 73) and ML n AF3-PLIP micro-F1 0.3458 (TP 204, FP 607, FN 165). Its paired protein-bootstrap delta was +0.2134, 95% interval [0.1700, 0.2549]. Its AF3-only PLIP contact F1 was 0.3070. This historical result was in the repository before the September run; it is not a score for September's 186-chain candidate training/test split.

There are three distinct '149' quantities that can otherwise be confused: the historical 149-protein OOF AF3/PLIP evaluation; the current September 149-chain training union; and the current September 37-chain held-out test union. The current 25-chain positive-only geometric-contact result (0.313 subtype; 0.297 REE) must **not** be compared as if it were the same experiment as the historical 149-protein PLIP result.

10. Interpretive limits and methodological risks

- **Positive-only selection:** twelve all-zero-label test chains were excluded from AF3. The raw combined REE model made 434 false-positive calls on those 12 (1,984 all-37 FP minus 1,550 positive-subset FP). Their AF3-assisted false-positive burden is unknown. Therefore no unbiased all-37 AF3-assisted F1 has been generated.
- **Contact definition mismatch:** September uses a permissive O/N/S distance ≤ 3.5 Å, while historical labels and the earlier AF3 result use PLIP metal-complex contacts. A geometric overlap can improve F1 but does not establish PLIP replication.
- **Ion-count leakage/conditioning:** HOLO ion count was derived from E1 subtype scores, APO contact smoothing and +2 padding. AF3 was asked to place the resulting ions, rather than independently deciding whether a protein binds. The 348 requested ions (up to 42 in one protein) could inflate geometric contact opportunities.
- **Element identification:** the experimental CIF supplied the metal identity. This is sensible for same-element structural validation, but it is extra information that a de novo unknown-target prediction would not have.
- **Structural confidence:** pTM concerns global topology, not correctness of individual ion coordinates or binding chemistry. No site-level pLDDT/PAE threshold or independent experimental coordinate matching was applied in this residue F1 analysis.
- **Label sparsity:** only 66 positive residues among 9,721 residues in the selected 25; specificity/TN can be high while precision remains modest. An unobserved contact is not necessarily a biologically certain negative.
- **No binding assay:** this is computational contact concordance, not measurement of affinity, selectivity between LREE/HREE, or experimentally confirmed binding.

11. Per-protein audit — all 25 completed HOLO cases

The first table gives all chain IDs, subtype, element, length, labeled positive residues, requested HOLO ion count, model pTM and per-job HOLO runtime. The next two tables give the per-residue confusion counts for raw E1 versus AF3-assisted predictions. All values are extracted from per_sample.csv and holo_receipt.tsv; aggregate counts were independently recomputed from the 25 rows before this PDF was built.

Chain	Type	Ion	Length	Truth +	Ions	pTM	HOLO s
1E8X_A	HREE	LU	841	4	27	0.69	791
1EXZ_B	LREE	SM	140	3	14	0.82	123
1GW6_A	HREE	YB	610	1	11	0.97	491
1K90_C	HREE	YB	503	2	16	0.68	487
1KQV_A	LREE	LA	75	3	8	0.83	81
1M9S_A	HREE	TB	523	3	13	0.60	489
1XR5_A	LREE	SM	460	3	21	0.93	271
2K0J_A	LREE	LA	147	5	17	0.65	125
4BD8_B	LREE	PR	152	2	9	0.85	125
4ESQ_A	HREE	TB	194	1	7	0.94	125
4XHA_B	HREE	LU	662	1	22	0.92	490
5AC2_A	HREE	YB	494	1	16	0.95	272
5AVK_A	LREE	PR	279	4	4	0.97	184
6FRM_H	HREE	TB	409	5	8	0.90	271
6HF7_A	HREE	TB	191	1	6	0.88	126
6OOM_A	LREE	PR	392	2	9	0.95	270
6TRT_A	HREE	TB	1369	5	42	0.85	1713
6ZV9_B	HREE	TB	169	2	14	0.67	125
7BXF_A	LREE	PR	369	3	24	0.82	273
8AB2_A	HREE	TB	329	2	8	0.90	183
8F7N_A	HREE	YB	364	3	15	0.66	186

Chain	Type	Ion	Length	Truth +	Ions	pTM	HOLO s
8RKI_A	HREE	YB	278	3	9	0.70	184
9C8X_A	LREE	LA	60	3	6	0.85	83
9C8Z_A	HREE	EU	60	3	6	0.85	80
9US6_B	HREE	TB	651	1	16	0.86	495

11.1 Matched-subtype residue results

TP/FP/FN cells show true positives, false positives and false negatives, in that order.

Chain	Type	Raw TP/FP/FN	Raw F1	Assisted TP/FP/FN	Assisted F1	Δ F1
1E8X_A	HREE	3/94/1	0.059	3/7/1	0.429	+0.369
1EXZ_B	LREE	2/24/1	0.138	2/12/1	0.235	+0.097
1GW6_A	HREE	0/32/1	0.000	0/3/1	0.000	+0.000
1K90_C	HREE	2/31/0	0.114	2/6/0	0.400	+0.286
1KQV_A	LREE	3/22/0	0.214	3/9/0	0.400	+0.186
1M9S_A	HREE	2/63/1	0.059	2/9/1	0.286	+0.227
1XR5_A	LREE	3/63/0	0.087	3/11/0	0.353	+0.266
2K0J_A	LREE	5/52/0	0.161	5/28/0	0.263	+0.102
4BD8_B	LREE	2/20/0	0.167	2/5/0	0.444	+0.278
4ESQ_A	HREE	0/15/1	0.000	0/3/1	0.000	+0.000
4XHA_B	HREE	1/79/0	0.025	1/11/0	0.154	+0.129
5AC2_A	HREE	0/52/1	0.000	0/2/1	0.000	+0.000
5AVK_A	LREE	1/22/3	0.074	1/0/3	0.400	+0.326
6FRM_H	HREE	5/25/0	0.286	5/5/0	0.667	+0.381
6HF7_A	HREE	0/20/1	0.000	0/2/1	0.000	+0.000
6OOM_A	LREE	2/39/0	0.093	2/4/0	0.500	+0.407
6TRT_A	HREE	5/134/0	0.069	4/14/1	0.348	+0.278
6ZV9_B	HREE	2/47/0	0.078	2/11/0	0.267	+0.188
7BXF_A	LREE	3/93/0	0.061	2/18/1	0.174	+0.113
8AB2_A	HREE	1/30/1	0.061	1/2/1	0.400	+0.339
8F7N_A	HREE	3/53/0	0.102	2/8/1	0.308	+0.206
8RKI_A	HREE	0/24/3	0.000	0/5/3	0.000	+0.000
9C8X_A	LREE	3/16/0	0.273	3/4/0	0.600	+0.327
9C8Z_A	HREE	3/14/0	0.300	3/4/0	0.600	+0.300
9US6_B	HREE	1/55/0	0.035	0/10/1	0.000	-0.035

11.2 Combined REE residue results

Chain	Type	Raw TP/FP/FN	Raw F1	Assisted TP/FP/FN	Assisted F1	Δ F1
1E8X_A	HREE	4/150/0	0.051	3/10/1	0.353	+0.302
1EXZ_B	LREE	2/23/1	0.143	2/12/1	0.235	+0.092
1GW6_A	HREE	1/56/0	0.034	1/7/0	0.222	+0.188
1K90_C	HREE	2/59/0	0.063	2/8/0	0.333	+0.270
1KQV_A	LREE	3/25/0	0.194	3/10/0	0.375	+0.181
1M9S_A	HREE	2/74/1	0.051	2/10/1	0.267	+0.216
1XR5_A	LREE	3/65/0	0.085	3/13/0	0.316	+0.231
2K0J_A	LREE	5/86/0	0.104	5/29/0	0.256	+0.152
4BD8_B	LREE	2/29/0	0.121	2/6/0	0.400	+0.279
4ESQ_A	HREE	0/21/1	0.000	0/5/1	0.000	+0.000
4XHA_B	HREE	1/121/0	0.016	1/10/0	0.167	+0.150
5AC2_A	HREE	0/59/1	0.000	0/5/1	0.000	+0.000
5AVK_A	LREE	1/24/3	0.069	1/0/3	0.400	+0.331

Chain	Type	Raw TP/FP/FN	Raw F1	Assisted TP/FP/FN	Assisted F1	Δ F1
6FRM_H	HREE	5/51/0	0.164	5/7/0	0.588	+0.424
6HF7_A	HREE	1/35/0	0.054	0/2/1	0.000	-0.054
6OOM_A	LREE	2/39/0	0.093	2/3/0	0.571	+0.478
6TRT_A	HREE	5/200/0	0.048	4/21/1	0.267	+0.219
6ZV9_B	HREE	2/41/0	0.089	2/12/0	0.250	+0.161
7BXF_A	LREE	3/95/0	0.059	2/19/1	0.167	+0.107
8AB2_A	HREE	2/46/0	0.080	2/4/0	0.500	+0.420
8F7N_A	HREE	3/105/0	0.054	2/10/1	0.267	+0.213
8RKI_A	HREE	2/41/1	0.087	1/5/2	0.222	+0.135
9C8X_A	LREE	3/21/0	0.222	3/4/0	0.600	+0.378
9C8Z_A	HREE	3/21/0	0.222	3/4/0	0.600	+0.378
9US6_B	HREE	1/63/0	0.031	0/11/1	0.000	-0.031

Per-protein F1 values are descriptive only; the headline F1 is pooled micro-F1, not the mean of the displayed rows. The machine-readable CSV also contains AF3-contact-only confusion counts and TN, precision and recall for every method and case.

12. Provenance, reproducibility and outstanding work

Artifact	Location / meaning
Current result	results/af3_assisted_2026-09-12/summary.json; per_sample.csv; holo_receipt.tsv; test_metrics.yaml
Current analysis code	heldout_af3_validation.py and run_heldout_af3_followup.sh at workspace root; run_af3_json_batch.sh
Current train output (remote)	/home/aid1/biology-stack/E1-Metal-Binding/runs/prototype_residue_level/2026-09-12-17-00
Current AF3 output (remote)	/home/aid1/biology-data/e1_validation_2026-09-12/{af3_apo_outputs,af3_holo_outputs,analysis,logs}
Current data candidate	E1-Metal-Binding-main/data/dataset_artifacts/ree_plip_2026-05-12_qa_rebuild
Historical PLIP report	E1-Metal-Binding-main/docs/af3/phase1_baseline_run_report_2026-04-04.md (May 17/20 updates)
Dataset audit	E1-Metal-Binding-main/docs/workflows/ree_neg020_canonical_dataset.md
Smoke input	biology-data/af3_input/1cnt_2.json; protein-only 130-aa AF3 installation test

To obtain a defensible full held-out score, run the 12 omitted all-zero-label chains through the same APO→HOLO protocol without using their labels to tune settings; include all 37 in the same primary denominator. Then reprocess all predicted HOLO models with PLIP under the training-label convention, compare geometric versus PLIP calls, inspect requested ion counts and predicted metal sites against experimental structures, and evaluate thresholds on a genuinely independent set. The current 25-case result should remain labeled 'positive-only, geometry-based diagnostic' until those steps are complete.

Finally, if this workflow becomes an LLM-plus-tools agent, the minimal scientific input unit is one chain sequence with an explicit target element or LREE/HREE class, immutable model/checkpoint and threshold IDs, MSA/embedding provenance, and optional experimental structure only for evaluation. The agent should call coarse, auditable tools for prepare→train/infer→APO→ion-count→HOLO→contact analysis→evaluation, recording a receipt and QC at each step. It should not infer ground-truth labels or claim binding from pTM alone.